

Treasury Forecast Tracker

A Microsoft Fabric and Power BI demonstration | Brett McDonald | 8 October 2026

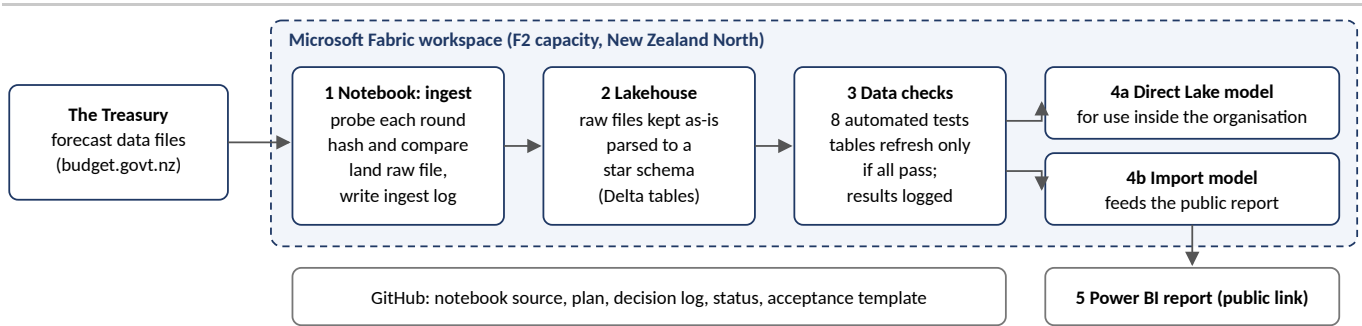
Live report (no sign-in): [open the Power BI report](#) | Code and project documents: github.com/brett-mcdonald-projects/treasury-forecast-tracker

What it does

The report answers one question: **how has the economic and fiscal outlook changed across the Treasury's successive forecast rounds?** It loads the Treasury's published Economic and Fiscal Update data files for three rounds (Half Year Update 2025, Budget Update 2026 and Pre-election Update 2026) and lets the reader compare any of eight headline measures across them, such as real GDP growth, CPI inflation, the operating balance and net core Crown debt.

I built it to show how I approach a reporting solution on Microsoft Fabric end to end: automated ingestion, a tested data model, and a governed route from source file to published report. It is an independent demonstration on public data and is not a Treasury product.

How it works



What is built and working today

Component	Detail
Source registry	One row per forecast round with its file address. A new round or a new source is a new row, not a redesign. Four rounds are registered, including Half Year Update 2026, which is not yet published.
Change detection	Each run requests every registered file, hashes it (SHA-256) and compares it with the last loaded hash in the ingest log. Only new or changed files are landed and processed. A missing file is recorded, not treated as a failure.
Lakehouse	Raw files are kept untouched in a timestamped landing folder. Parsed data is held in eight Delta tables: 12,411 observations across 48 Treasury series and three rounds, 1,421 rows of derived headline measures, plus the registry, ingest log and test results.
Data checks	Eight tests run before any table is refreshed: row presence, duplicate keys, nulls, referential integrity, required series present in every round, no gaps in the quarterly GDP series, forecast flag complete, and a plausibility range. A failed test stops the refresh.
Semantic models	A Direct Lake model over the lakehouse, and an import-mode copy for the public report, because Power BI's public publishing does not support Direct Lake.
Report	Two pages. <i>Forecast tracker</i> : measure selector, chart and table by round. <i>Pipeline and data quality</i> : the ingest log and test results, so the reader can see what the pipeline did.

Result of the run on 8 October 2026: three rounds fetched directly from budget.govt.nz and loaded (HTTP 200); Half Year Update 2026 returned 404 and was logged as not yet published; 8 of 8 data checks passed. When the Treasury publishes that file under its usual naming, the next run will pick it up without a code change.

Design choices

- **Definitions are computed once, in the data layer.** Growth rates and percent-of-GDP measures are derived in the notebook, so every report built on the model uses the same calculation.
- **Raw data is never overwritten.** Each fetched file is stored as received, so any figure can be traced back to the file it came from.
- **The pipeline reports on itself.** The ingest log and test results are tables in the model and are shown in the report.
- **Neutral presentation.** The report describes how forecasts moved between rounds. It does not assess forecast quality.
- **One documented heuristic.** The economic data file has no forecast flag, so the last quarter with a whole-number CPI index is treated as the last actual. The fiscal file carries its own flag, which is used directly.

What is not built yet

This was built in under a day as a first release. The following are planned and tracked in the repository, and are not claimed as done:

Item	Status
Scheduled run	The notebook is run on demand today. Next step is a scheduled Fabric pipeline with a failure alert.
Dev, Test, Prod	One workspace so far. Next step is Fabric Git integration and a deployment pipeline with an acceptance checklist before release.
Reconciliation	Tests check structure and integrity. A check of selected figures against the Treasury's published tables is still to add.
Actuals	Stats NZ outturns (GDP, CPI, unemployment) as a second source, to show actual against forecast.
Report polish	Round order in the legend is alphabetical, not chronological; field names are shown as stored; no reusable theme file yet.

How it was built

I set the requirements (one cloud platform, automated ingestion from a predictable public source, change detection, an architecture that can take further sources, and a full release life cycle rather than a one-off build) and chose the subject and approach. I directed an AI assistant (Claude), which wrote the notebook code and carried out much of the Fabric configuration, and I reviewed the results. The decision log and status file in the repository record what was decided and why.

Platform

Microsoft Fabric (lakehouse, PySpark notebook, Direct Lake and import semantic models), Power BI service, GitHub. Built entirely in the browser. Data: the Treasury, Economic and Fiscal Update data files (budget.govt.nz).

The report

The two pages that follow are exported from the live Power BI report.

Treasury Forecast Tracker

- measure
- ☐

Core Crown expenses, % of...
- ☐

Core Crown tax revenue, % ...
- ☐

CPI inflation
- ☒

Net core Crown debt, % of ...
- ☐

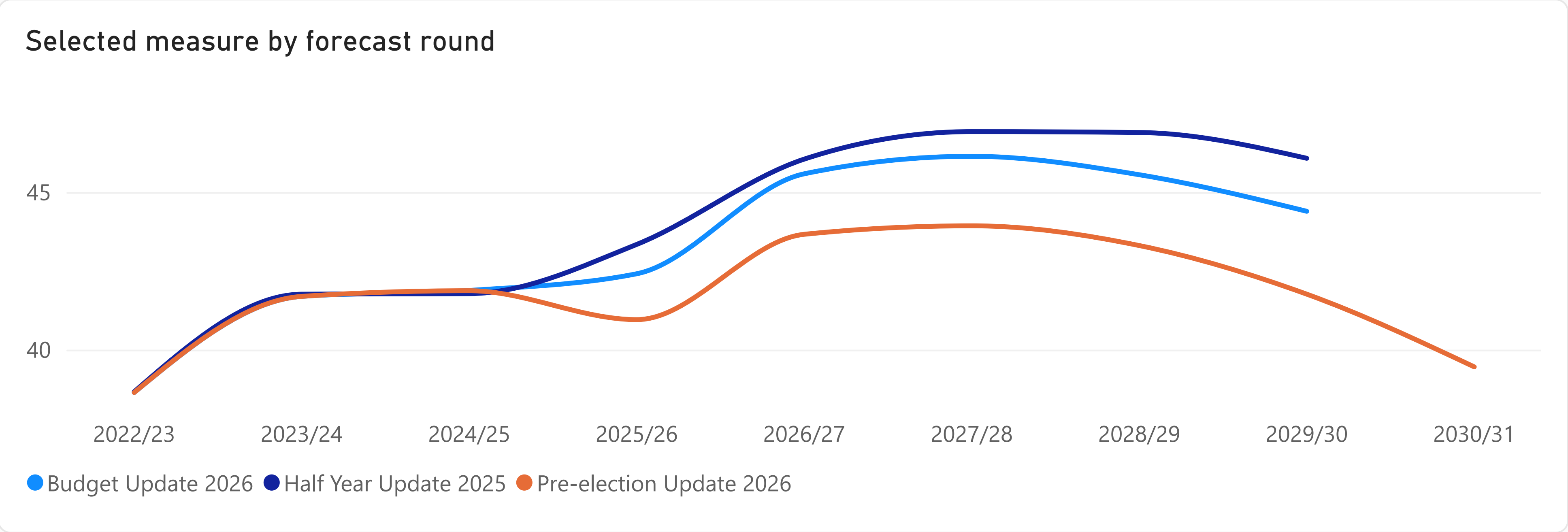
OBEGAL (excluding ACC)
- ☐

OBEGAL (excluding ACC), %...
- ☐

Real GDP growth
- ☐

Unemployment rate

How the economic and fiscal outlook changed across three Treasury forecast rounds. Pick a measure above. Source: the Treasury, Economic and Fiscal Update data files (budget.govt.nz), loaded by an automated Microsoft Fabric pipeline. Independent demonstration by Brett McDonald; not a Treasury product.



period_label	Budget Update 2026	Half Year Update 2025	Pre-election Update 2026
2022/23	38.64	38.67	38.64
2023/24	41.69	41.76	41.69
2024/25	41.88	41.78	41.87
2025/26	42.41	43.34	40.95
2026/27	45.58	46.05	43.66
2027/28	46.14	46.92	43.93
2028/29	45.55	46.89	43.31

Ingestion log: sources probed on the latest pipeline run (a 404 means the round is not yet published)

round_code	status	Sum of http_status	Sum of bytes	url
BEFU26	loaded	200.00	48,676.00	https://budget.govt.nz/budget/excel/befu2026/befu26-economic-forecasts-data.xlsx
HYEFU25	loaded	200.00	110,564.00	https://budget.govt.nz/budget/excel/hyefu2025/hyefu25-economic-forecasts-data.xlsx
HYEFU26	not_published	404.00		https://budget.govt.nz/budget/excel/hyefu2026/hyefu26-economic-forecasts-data.xlsx
PREFU26	loaded	200.00	60,712.00	https://budget.govt.nz/budget/excel/prefu2026/prefu26-economic-forecasts-data.xlsx

Automated data checks: all must pass before the reporting tables refresh

test	passed	detail
every fact series is in dim_series	True	
fact has rows	True	12411
forecast flag set on every row	True	
headline source series present in every round	True	['lhurzq', 'ngdpp_zq', 'nzgpfi0024', 'nzgpfiobegalexacc', 'nznaac0317_fisc', 'pcpiq']
no duplicate fact keys	True	